

f, a and b are connected to port pins 2.0 through P2.7, respectively. This is actually a trade-off as the simplicity in the PCB layout has increased the complexity of the scanning software.

Port P2 provides the segment data for 7-segment displays DIS1 through DIS6. Port pins P0.0 through P0.7 are pulled up with 10-kilo-ohm resistor network (RNW1) and used to select the seven-segment displays. These pins drive the base pins of pnp transistors T3 through T8 to select the corresponding display. For example, if pin P0.2 goes low, transistor



Fig. 2: Pin configurations of 7805 and BC557

T3 is driven into saturation, providing supply to the common-anode pin 3 or 8 of 7-segment display DIS1.

Microcontroller AT89C52 provides the segment data and display-enable signal simultaneously in time-division multiplexing mode for displaying a particular number on a seven-segment display. Segment data and display-enable pulse for the display is refreshed continuously. Thus the display appears to be continuous even though the displays light up one by one.

Switches S1 through S4 are used to set the date, time and alarm. Port pins P3.0 through P3.3 are pulled up by resistors R1 through R4 and connected to switches S1 through S4, respectively. Switch S5 is used to manually reset the

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

microcontroller, while the power-on reset signal for the microcontroller is derived from the combination of capacitor C1 and resistor R5.

A 12MHz crystal is used to generate the clock frequency for the microcontroller. Port pin P0.1 of the microcontroller drives LED1 to provide dot between DIS4 and DIS5. It also drives LED2 to provide dot between DIS2 and DIS3. Port pin P0.0 drives piezobuzzer PZ1 for sounding the alarm.

To derive the power supply, 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 12V, 500 mA. The transformer output is rectified by a full-wave bridge rectifier comprising diodes D1 through D4, filtered by capacitor C4 and regulated by IC 7805 (IC2). Capacitor C5 bypasses the ripples present in the regulated output. LED3 acts as the power indicator.

Resistor R22 limits the current through LED3. A rechargeable battery is used for power backup.

Construction and working

An actual-size, single-side PCB for the real-time clock is shown in Fig. 4 and its component layout in Fig. 5. Assemble the circuit on a PCB to minimise time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Use proper IC base for the microcontroller. Before inserting the IC, check all the supply voltages. When power is switched on, the display should show '00.00.00'. It will show 01.01.01 when any key is pressed before setting.

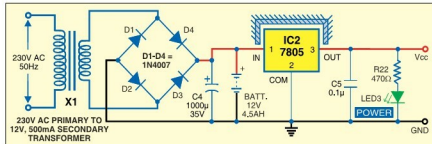


Fig. 3: Power supply circuit

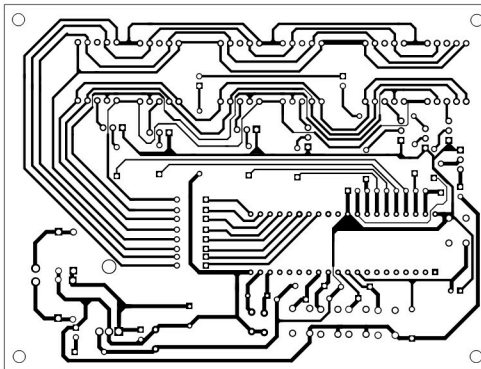


Fig. 4: An actual-size, single-side PCB for the real-time clock